← Back to O5 Algebra and inequalities

O5 Problems: algebra and inequalities

Thirty problems in three levels. The topic of each problem is given in italics. For Level 2 and Level 3, write a complete solution: for "find all" problems, prove that your list is complete *and* check that every item on it works; for inequalities, state when equality holds. Hints are at the end, one line per problem. Solutions are in solutions.md.

Level 1 — Kangoeroe style (wizPROF)

Multiple choice. Aim for about 5 minutes per problem, but write down *why* your answer is right.

1. *Algebraic manipulation.* The real number x satisfies $x + \dfrac{1}{x} = 3$. What is $x^3 + \dfrac{1}{x^3}$?

(A) 15 (B) 18 (C) 21 (D) 24 (E) 27

2. *Factoring identities.* What is the value of

$$\frac{2026^3 - 2025^3 - 1}{2025 \cdot 2026}\,?$$

(A) 1 (B) 2 (C) 3 (D) 2025 (E) 4051

3. *Remainder theorem.* What is the remainder when $x^{2030} + 2x + 3$ is divided by $x + 1$?

(A) 0 (B) 1 (C) 2 (D) 4 (E) 6

4. *Vieta's formulas.* The equation $x^2 - 7x + 3 = 0$ has roots a and b. What is $a^2 + b^2$?

(A) 37 (B) 43 (C) 46 (D) 49 (E) 55

5. *Telescoping.* What is the value of

$$\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \frac{1}{\sqrt{99}+\sqrt{100}}\,?$$

(A) 9 (B) 10 (C) $\sqrt{99}$ (D) 99 (E) $\frac{99}{10}$

6. *Arithmetic sequence.* In an arithmetic sequence (rekenkundige rij) the third term is 10 and the tenth term is 31. What is the sum of the first 20 terms?

(A) 610 (B) 630 (C) 650 (D) 670 (E) 690

7. *AM-GM.* What is the smallest possible value of $\dfrac{x^2 + 9}{x}$ for positive real x?

(A) 3 (B) 6 (C) 9 (D) 10 (E) there is no smallest value

8. *Functional equation.* A function f is defined for all nonzero real numbers and satisfies $f(x) + 2f\left(\dfrac{1}{x}\right) = 3x$ for every $x \neq 0$. What is $f(2)$?

(A) -1 (B) 0 (C) 1 (D) 2 (E) 3

Level 2 — NWO first/second round style

Short answers with a full justification. About 20–30 minutes per problem.

9. *Sophie Germain identity.* Find all positive integers n for which $n^4 + 4$ is a prime number.

10. *Symmetric expressions, Newton sums.* Real numbers a, b, c satisfy $a+b+c = 6$, $a^2+b^2+c^2 = 14$ and $a^3 + b^3 + c^3 = 36$. Find abc and $a^4 + b^4 + c^4$.

11. *Integer roots, Vieta.* Find all integers a for which both roots of $x^2 + ax + 2a = 0$ are integers. (A double root counts, as two equal integer roots.)

12. *Remainder theorem.* A polynomial $P(x)$ leaves remainder 3 when divided by $x - 1$, and remainder 5 when divided by $x-2$. What is the remainder when $P(x)$ is divided by $(x-1)(x-2)$?

13. *Recurrence, induction.* The sequence $a_1, a_2, \ldots$ is given by $a_1 = 3$ and $a_{n+1} = a_n^2 - 2a_n + 2$.

(a) Find a closed formula for a_n.

(b) Prove that $a_1 a_2 \cdots a_n = a_{n+1} - 2$ for every $n \geq 1$.

(c) Deduce that any two different terms of the sequence are coprime (relatief priem).

14. *Telescoping.* Find a closed formula for

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)}.$$

15. *AM-GM.* Prove that $(a + b)(b + c)(c + a) \geq 8abc$ for all positive reals a, b, c. When does equality hold?

16. *Cauchy–Schwarz, Engel form.* Positive reals x, y, z satisfy $x + y + z = 1$. Prove that

$$\frac{1}{x} + \frac{4}{y} + \frac{9}{z} \geq 36,$$

and find all cases of equality.

17. *Cauchy–Schwarz, Engel form.* Prove Nesbitt's inequality (ongelijkheid van Nesbitt): for positive reals a, b, c,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

When does equality hold?

18. *Rearrangement, AM-GM.* Prove that $a^3+b^3+c^3 \geq a^2b+b^2c+c^2a$ for all positive reals a, b, c, and find the equality cases.

19. *Cauchy–Schwarz.* Real numbers x, y satisfy $3x + 4y = 10$. Find the smallest possible value of $x^2 + y^2$, and the pair (x, y) where it is reached.

20. *Functional equation.* Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) + 2f(1 - x) = x^2$ for all real x.

21. *Functional equation.* Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x)f(y) - f(xy) = x + y$ for all real x, y.

Level 3 — NWO final / BxMO style

Full proofs. Allow 45–90 minutes per problem, possibly spread over two sessions.

22. *Three-cube identity, number theory.* Determine all integers n that can be written as $n = a^3 + b^3 + c^3 - 3abc$ with a, b, c integers.

23. *Vieta, substitution.* Let a, b, c be the three (complex) roots of $x^3 - x - 1 = 0$. Compute

$$\frac{1+a}{1-a} + \frac{1+b}{1-b} + \frac{1+c}{1-c}.$$

24. *AM-GM, QM-AM.* Positive reals a, b, c satisfy $abc = 1$. Prove that $a^2 + b^2 + c^2 \geq a + b + c$, and find the equality cases.

25. *Tangent line trick.* Positive reals a, b, c satisfy $a + b + c = 1$. Prove that

$$\frac{a}{1+a^2} + \frac{b}{1+b^2} + \frac{c}{1+c^2} \leq \frac{9}{10}.$$

26. *Substitution, Engel form, AM-GM.* Positive reals a, b, c satisfy $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

27. *Sequences, bounding.* The sequence $a_1, a_2, \ldots$ is given by $a_1 = 1$ and $a_{n+1} = a_n + \dfrac{1}{a_n}$. Prove that $14 < a_{100} < 15$.

28. *Polynomials, comparing coefficients.* Find all polynomials P with real coefficients such that $P(x^2) = P(x)^2$ for all real x.

29. *Cauchy's equation over $\mathbb{Q}$.* Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f(x+y) = f(x) + f(y) + xy$ for all rational x, y.

30. *Functional equation, injectivity, surjectivity.* Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + f(y)) = y + f(x)^2$$

for all real x, y.

Hints

1. Cube both sides of $x + \frac{1}{x} = 3$.
2. Put $a = 2026$, $b = -2025$, $c = -1$; what is $a + b + c$?
3. Use the remainder theorem with $x = -1$.
4. $a^2 + b^2 = (a+b)^2 - 2ab$.
5. Multiply each fraction by $\sqrt{k+1} - \sqrt{k}$ over itself.
6. Find the common difference d from the two given terms.
7. Split the fraction as $x + \frac{9}{x}$.
8. Also substitute $x = \frac{1}{2}$ and solve two equations in two unknowns.
9. Write $n^4 + 4$ as a difference of two squares.
10. First find $ab + bc + ca$, then use the three-cube identity or Newton's sums.

11. If the roots are r, s, Vieta gives $rs = -2(r+s)$; add 4 to both sides and factor.
12. The remainder is $ax + b$; plug in $x = 1$ and $x = 2$.
13. Look at $a_n - 1$; then use $a_{n+1} - 2 = a_n(a_n - 2)$.
14. Write $\frac{1}{k(k+1)(k+2)}$ as $\frac{1}{2}$ times a difference of two fractions.
15. Use AM-GM on each bracket.
16. Write the numerators as $1^2, 2^2, 3^2$ and use the Engel form.
17. Write $\frac{a}{b+c} = \frac{a^2}{a(b+c)}$.
18. Apply AM-GM to a^3, a^3, b^3.
19. Use Cauchy–Schwarz with the vectors $(3, 4)$ and (x, y).
20. Replace x by $1 - x$ and solve the system for $f(x)$.
21. Find $f(0)$ first, then put $y = 0$.
22. Write the expression as $s(s^2 - 3q)$ with $s = a + b + c$, $q = ab + bc + ca$, and look modulo 3 and 9.
23. Put $y = \frac{1+x}{1-x}$, solve for x, and find the cubic equation satisfied by y.
24. First show $a + b + c \geq 3$; then use QM-AM.
25. Find the tangent line of $f(x) = \frac{x}{1+x^2}$ at $x = \frac{1}{3}$ and show f lies below it.
26. Substitute $x = \frac{1}{a}$, $y = \frac{1}{b}$, $z = \frac{1}{c}$.
27. Study a_n^2; for the upper bound, compare $\sum \frac{1}{k}$ with blocks between powers of 2.
28. If P is not of the form cx^n, look at the two highest nonzero terms.
29. Subtract $\frac{x^2}{2}$ from f.
30. Put $x = 0$ to get bijectivity, then show $f(0) = 0$, $f(x^2) = f(x)^2$, and that f is increasing.